

Chuong Dang TA

📍 Stockholm, Sweden | Willing to relocate

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🔗 chuongta.github.io

About Me

Energy systems and data science enthusiast with a thermal engineering background. Expected to graduate in **September 2026** from an Erasmus Mundus MSc (**DENSYS**). For my **Master's Thesis** at **rebase.energy** (Stockholm), I forecasted wind turbine icing power losses using an extended **IEA Wind Task 19** framework and **LightGBM**. Active builder of energy data side projects, including probabilistic day-ahead price forecasting, smart heat pump control, and energy storage optimization. Published author in 2 peer-reviewed Q1 journals (IFs: 11.8, 9.2).

Education and Training

University of Lorraine, France & Politechnique of Turin, Italy - densys.univ-lorraine.fr *Sep 2024 – Sep 2026*
Master of Science in Decentralised and Smart Energy Systems (DENSYS)

A two-year Erasmus Mundus Joint Master's program (120 ECTS) focusing on the modeling, control, and optimization of decentralised and renewable energy systems.

Selected Coursework:

- University of Lorraine: [Data and Forecasting in Microgrids], [Optimal Local Design Energy Network].
- Politecnico di Turin: [Polygeneration and Advanced Energy Systems], [Smart Electricity Systems].

Hanoi University of Science and Technology (HUST), Vietnam - hust.edu.vn/en *Aug 2018 – Jan 2023*
Bachelor in Thermal Engineering

Graduated **1st/218** with a CGPA of 3.54/4.00 from Vietnam's leading technical university. Core studies covered thermodynamics, heat transfer, district heating and cooling systems, and power plant systems.

Bachelor Thesis: “*Techno-Economic Analysis of Internal Combustion Engines Using Diesel and LNG*” (Grade: 9.5/10)

Scientific Research - **ORCID: 0009-0009-7530-7430**

- Dang-Chuong Ta; Thanh-Hoang Le; Long Van Phan; Hoang-Luong Pham, “Feasibility Analysis of Hydrogen Co-Firing in Vietnam's Gas Power Plants for the Period 2035–2050,” *Energy Conversion and Management*, Impact Factor: 11.8, **2025**. DOI: [10.1016/j.enconman.2025.120192](https://doi.org/10.1016/j.enconman.2025.120192)
- Dang-Chuong Ta; Thanh-Hoang Le; Hoang-Luong Pham, “An Assessment of the Potential for Large-Scale Hydrogen Export from Vietnam to Asian Countries: Techno-Economic Analysis, Transport Options, and Energy Carrier Comparison,” *International Journal of Hydrogen Energy*, Impact Factor: 9.2, **2024**. DOI: [10.1016/j.ijhydene.2024.04.033](https://doi.org/10.1016/j.ijhydene.2024.04.033)
- (Ongoing) Dang-Chuong Ta et al., “Wind Farm-Scale Forecasting of Turbine Icing Power Losses Using Multi-Ridge Regression Models,” based on MSc thesis research at rebase.energy - extending the turbine-level classifier-regressor framework with multi-ridge regression models and upscaling to wind farm-level forecasts.

Work Experience

rebase.energy, Stockholm, Sweden

Jul 2026 – Present

Junior Data Scientist - Wave Energy Converter Power Forecasting

- **WEC Power Forecasting:** Autoregressive power forecasting via a **Multi-Ridge Regression** stack (**Elastic Net** and **LightGBM**).
- **Wave Energy Flux Forecasting:** Forecasting wave energy flux from **ERA5 Ocean Wave** data, benchmarking regularised and tree-based models (**Elastic Net**, **Random Forest**, **LightGBM**) against probabilistic methods (**Quantile Regression**, **Quantile Regression Forests**, **Bootstrapped residuals**).
- **Skills Learned:** Multi-Ridge Regression stacking, probabilistic forecasting, and working with ERA5 ocean wave reanalysis data.

- **Icing Detection & Feature Engineering:** Labeled icing events on multi-year Swedish turbine SCADA data via an extended **IEA Wind Task 19 Sigmoid Performance Ratio** framework, corrected SCADA-ERA5-Land biases with **Quantile Mapping**, and used **Spearman/SHAP** screening to select stable predictors.
- **Icing Power-Loss Forecasting:** Built a two-stage **LightGBM** classifier-regressor with walk-forward cross-validation to forecast icing losses at **1-36 hour** lead times, showing SCADA signals dominate near-term skill while NWP forecasts matter more for longer horizons.
- **Skills Learned:** Time series forecasting under non-stationarity, and bias correction of NWP data against ground-truth SCADA.

Selected Projects - chuongta.github.io/EnergyForecasting

- **LLM Assistant for Operational BESS Dispatch**
Integrated LightGBM day-ahead price forecasting with a Pyomo scheduling model, using a local LLM agent to query and explain optimal battery dispatch schedules.
- **Probabilistic Electricity Price Forecasting**
Developed day-ahead price forecasting models (Quantile Regression, Quantile Regression Forests, and bootstrapped residuals) on Danish DK1 data, evaluated with walk-forward validation.
- **Smart Heat Pump Control**
Built a physical model and Gradient Boosting digital twin of a heat pump, using Reinforcement Learning (PPO) to optimize thermal storage dispatch for price arbitrage.
- **Stochastic Storage Optimisation**
Developed a two-stage stochastic battery scheduling model in Pyomo, using probabilistic forecasts to optimize dispatch under price uncertainty.

Technical Skills

- **Machine Learning:** Supervised, unsupervised, and reinforcement learning; hybrid, physics-informed models for energy system forecasting and predictive maintenance.
- **Data Science & Optimisation:** Applied **MILP, MINLP, and heuristic algorithms** (e.g., genetic algorithms) in Pyomo to optimize renewable energy systems.
- **Technical Toolset:** Python (PyTorch, Scikit-learn, TensorFlow, Pyomo, PyWake), Ollama (local LLM), Git, Julia, MATLAB/Simulink, Modelica, QBlade, HOMER Pro, GIS, Aspen Plus, Typst, LaTeX.

Honors & Awards

- **Erasmus Mundus scholarship** for DENSYS Joint Master Program
- **JNED Award** for research on Japan's nuclear facilities (2023)

Short Courses & Certifications

- **Self-Directed Learning:** DTU course **Renewables in Electricity Markets (46755)** ([Personal Notes](#)). Topics: market structures, pricing mechanisms, day-ahead clearing, intraday trading, balancing markets, ancillary services, stochastic bidding.
- Optimisation with Python: Complete Pyomo with Bootcamp A-Z

Hobbies & Interests

🏹 **STRAVA** [Chương Đăng Tà](#) | [Strava Profile](#)

Active Training

🏊 Swimming 🚴 Cycling 🏃 Running. Aspiring to complete a triathlon!

Language

- Vietnamese (Native)
- English (IELTS 7.5 - Full Professional)
- German (A2), French (A1)

Referees

Prof. Fabrice Lemoine - Chair of the DENSYS Erasmus Mundus Joint Master Degree, Université de Lorraine

✉ fabrice.lemoine@univ-lorraine.fr

Marta Gandiglio - Associate Professor, Department of Energy (DENERG), Politecnico di Torino

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